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EGCPPIS: learning hierarchical equivariant graph representations with contrastive integration for protein–protein interaction site identification

Guicong Sun¹, Yongxian Fan^{1*}, Yangfeng Zhu¹ and Mengxin Zheng¹

*Correspondence:

Yongxian Fan

yongxian.fan@gmail.com

¹School of Computer Science
and Information Security, Guilin
University of Electronic Technology,
Guilin 541004, China

Abstract

Background Protein–protein interactions regulate the dynamic operation of intracellular molecular networks, serving as the molecular basis for revealing protein functions and disease mechanisms. Recently, several computational methods for predicting protein–protein interaction sites (PPIs) have been presented as alternatives to costly and labor-intensive traditional experiments. However, existing methods generally ignore the inherent hierarchical structure of protein chains. Furthermore, the equivariance of graph structure during spatial transformations is often neglected when applying graph neural networks to modeling. Therefore, accurately identifying PPIs remains a challenging task.

Results In this work, we propose an end-to-end GNN-based computational method, EGCPPIS, for efficiently identifying protein–protein interaction sites. First, we construct a hierarchical graph representation of the protein chain, including residue-level graph and atom-level graph. Next, EGCPPIS designs an E(n) Equivariant Graph Neural Network (EGNN) module to learn residue-level embeddings with equivariant features. After further extracting atom-level embeddings using the GraphSAGE module, we introduce the contrastive learning strategy to integrate hierarchical graph features. This strategy enables us to learn consistent embeddings between residue-level and atom-level representations. Finally, the fused embeddings are weighted using an improved gated multi-head attention mechanism.

Conclusion Comprehensive evaluation results on multiple datasets demonstrate that EGCPPIS significantly outperforms state-of-the-art methods. Extensive comparative experiments and case studies further confirm that EGCPPIS can reveal the decision-making patterns in PPIs prediction, facilitating the discovery of potential PPIs. The original datasets and code of EGCPPIS are available at <https://github.com/GuicongSun/EGCPPIS>.

Keywords Protein–protein interaction site identification, E(n) equivariant graph neural network, GraphSAGE, Contrastive learning, Gated multi-head attention mechanism



Introduction

Proteins, as the core molecules of life activities, play crucial roles in various biological processes within living organisms [1–3]. Protein–protein interactions (PPI) are essential in regulating intracellular signal transduction [4, 5], metabolism and transport [6]. Therefore, accurately identifying protein–protein interaction sites (PPIs) is crucial for understanding protein functions and designing protein structures [7–9]. Traditional experimental techniques, such as yeast two-hybrid [10] and affinity purification-mass spectrometry [11], can provide high-confidence PPIs data, but are typically limited by high costs, long cycles, and restricted applicability. Thus, there is an urgent need for developing efficient computational methods to accurately identify PPI binding sites.

Referring to recent reviews, existing PPIs prediction methods can be divided into two categories: sequence-based methods and structure-based methods [12, 13]. Sequence-based methods utilize amino acid chains and evolutionary profiles, concentrating on the linear features within sequences [14]. These methods take sequence features as input and apply deep learning techniques for capturing hidden residue interaction patterns within sequences. For instance, DeepPPISP employs a sliding window strategy combined with text-based convolutional neural networks (CNNs) to mine local and global sequence attributes [15]. DELPHI introduces three novel feature types and integrates CNN and RNN modules to construct a fine-tuned ensemble model for PPIs prediction [16]. ProB-site generates three features sets from protein sequences and uses CNNs to extract latent local information. HN-PPISP designs a multi-branch hybrid module to learn local and global sequence features [17]. Seq-InSite incorporates two protein language models (pLMs) embeddings and employs a hybrid MLP-LSTM architecture for PPIs prediction [18]. EnsemPPIS introduces the transformer-based architecture to learn residue interactions in protein sequences and applies ensemble learning strategies to integrate local and global features [19]. Sequence-based methods are more generalizable and especially effective when the structural information of proteins is unavailable. It is worth noting that the advent of AlphaFold has provided credible solutions for protein structure prediction [20, 21]. In addition, the rapid development of graph neural networks (GNNs) has enabled more effective utilization of three-dimensional (3D) structural information of proteins [22, 23].

The performance improvement of structure-based methods benefits from the rich representation of protein structural features and the widespread application of GNNs. For instance, GraphPPIS applies the initial residual and identity mapping strategies to build deeper GNN models, enhancing model performance [24]. GHGPR-PPIS extends the generalized PageRank algorithm within a GNN framework to predict PPIs [25]. EGRET leverages transfer learning strategy and introduces an edge-aggregated network to further improve PPIs prediction performance [26]. AGAT-PPIS adds the representation of edge features and uses an augmented graph attention network for modeling [27]. HSSPPI generates atomic graphs to assist in protein structure representation and extracts multi-scale features using spatial-sequential blocks [28]. AGF-PPIS employs a multi-attention module for weighting sequence features and then transfers them to GNNs module to capture neighborhood residue information [29]. GACT-PPIS predicts PPIs using a deep transformer architecture and an augmented graph attention network as the basic unit [30].

Although computational methods for PPIs prediction have achieved significant progress, several challenges still remain. First, sequence-based methods ignore the topological structure of proteins and lack the ability to represent critical spatial features. In contrast, structure-based methods can utilize 3D spatial information to provide more accurate predictions. However, existing structure-based methods are generally based on residue graph modeling, lacking fine-grained atomic representations and ignoring the hierarchical interactions between the two levels of features. Additionally, structure-based methods generally use GNN and its variants to build models, but traditional GNN-based models lose the original spatial structure of proteins during the message-passing process. In contrast, E(n) Equivariant Graph Neural Networks (EGNN) [31] preserve spatial equivariance while aggregating node information and have been successfully applied in various protein modeling tasks [32–34]. For example, EquiPNAS constructed an equivariant deep graph neural network to improve protein-nucleic acid binding site prediction [32].

Given the limitations of existing methods, there is an urgent need to develop efficient PPI prediction methods. In this paper, we propose a novel deep learning framework called EGCPPIs, which identifies PPIs by fusing hierarchical graph consistency information through the graph contrastive learning strategy. First, we construct the corresponding hierarchical graph based on the protein sequence. Each residue in the residue-level graph corresponds to a set of atoms in the atom-level graph. Initial node embeddings in the residue-level graph are derived from the sequence and structural features of proteins, while atom-level node embeddings are initialized using binary encodings of atom types. EGCPPIs designs a dual-channel graph deep learning framework to capture different levels of structural information, respectively. The residue-level graph is modeled by the EGNN module, while the atom-level graph is modeled by the GraphSAGE module. Subsequently, we employ graph contrastive learning to align and integrate embeddings from the two graph levels, enhancing feature consistency across the hierarchy. Finally, an improved gated multi-head attention module updates fused residue embeddings, and a multi-layer perceptron (MLP) is used for final predictions. Comprehensive experimental results on two public datasets demonstrate that EGCPPIs significantly outperforms the state-of-the-art methods. Figure 1 illustrates the overall architecture of EGCPPIs.

Materials and methods

Benchmark datasets

To evaluate EGCPPIs's performance objectively, we utilized two datasets derived from previous PPIs prediction tasks proposed by DeepPPISP [15] and GraphPPIS [24], respectively. Table 1 records the details of the two benchmark datasets. Additionally, we visualize the distribution of protein sequence lengths for each dataset using violin plots in Fig. 2 (see Supplementary Table S1 for details).

Zeng et al. integrated three widely used benchmark datasets in their work on DeepPPISP, including Dset_186, Dset_72 [35], and PDBset_164 [36]. All three datasets were derived from RCSB Protein Data Bank (PDB) [37]. Sequences were filtered based on stringent criteria of resolution $< 3 \text{ \AA}$ and sequence identity $< 20\%$, resulting in a mixed dataset of 422 protein sequences. For consistency with previous work such as DeepPPISP, the dataset was similarly split into Train352 for training and Test70 for independent evaluation.

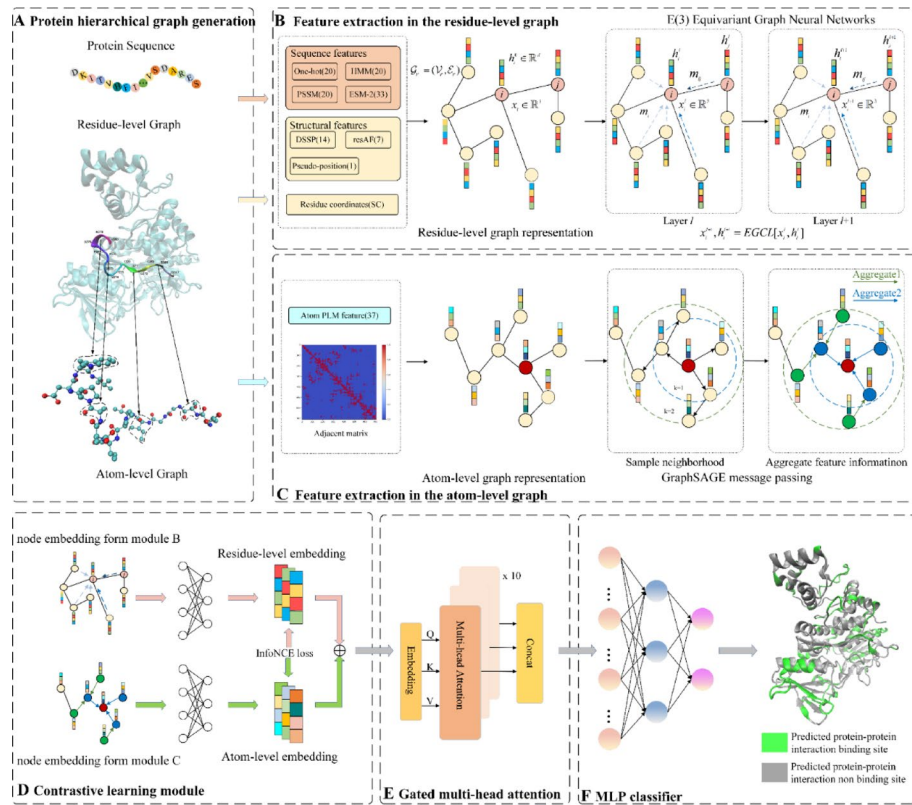


Fig. 1 Illustration of the EGCPPIs method for PPIs prediction. **A** Generating a hierarchical graph consistent with the protein sequence, including a residue-level graph and an atom-level graph. **B** The EGNN module takes the featurized residue-level graph representation as input to learn residue-level node embeddings. **C** The GraphSAGE module takes the featurized atom-level graph representation as input to learn atom-level node embeddings. **D** Integrating the embeddings of the residue-level graph and atom-level graph through the graph contrastive learning strategy. **E** Utilizing an improved gated multi-head attention mechanism to weight the residue node embeddings. **F** Performing residue-level PPIs prediction using the MLP classifier

Table 1 Details of benchmark datasets used in the two PPIs prediction tasks

Task	Dataset	No. proteins	No. binding residues	No. non-binding residues	% of binding residues
DeepPPISP	Train352	352	11,079	62,102	15.14
	Test70	70	2332	9459	19.78
GraphPPISP	Train_335-1	334	10,374	66,366	13.52
	Test_60	60	2075	11,069	15.79
	Test_315-28	287	8566	51,810	14.19
	Ubtest_31-6	25	711	5206	12.02

For the GraphPPIS task, four datasets were organized: Train_335, Test_60, Test_315, and UBtest_31. Similar to DeepPPISP, Train_335 and Test_60 were derived from the same three sub-datasets. Yuan et al. applied BLASTClust [38] to remove redundant sequences with >25% similarity and >90% overlap, resulting in 395 protein chains. Among them, 335 sequences were randomly selected for the training set (Train_335), with the remaining 60 sequences were used as an independent test set (Test_60). Test_315 was derived from protein complexes resolved in the PDB database. In addition, GraphPPIS constructed an unbound test set (UBtest_31) to explore the effect of conformational changes on PPIs prediction. Specifically, Yuan et al. retrieved 31 protein chains

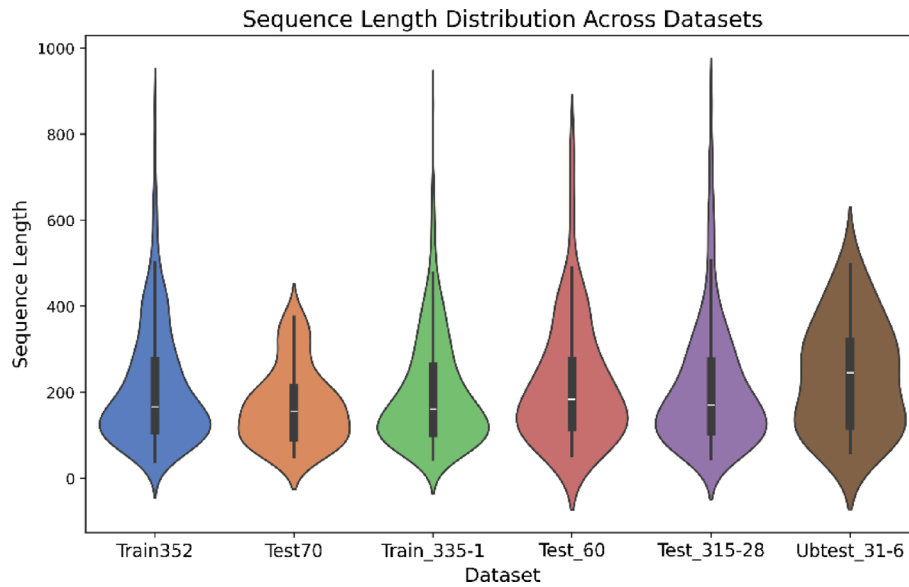


Fig. 2 Sequence length distributions of the benchmark datasets

with known unbound structures from the PDB that correspond to entries in the Test_60 dataset to construct UBtest_31. UBtest_31 is composed of 31 protein chains with known monomer structures retrieved from Test_60 on the PDB website. We fine-tuned four datasets for the GraphPPIS task, removing abnormal protein chains inconsistent with structures recorded in the PDB database. The final datasets used in this study include Train_335-1, Test_60, Test_315-28, and Ubtest_31-6.

Protein hierarchical graph construction

The raw data for PPIs prediction task is represented as a list of protein sequences. In this study, each protein is represented by a hierarchical graph that includes two levels: a residue-level graph and an atom-level graph. Residues serve as nodes in the residue-level graph, and each residue is mapped to a set of atoms in the atom-level graph. The 3D conformation of proteins shows its inherent hierarchy. To this end, we introduce the atom-level graph, which is often ignored in traditional studies, to achieve a more fine-grained residue representation [39].

The residue-level graph of each protein monomer can be defined as $\mathcal{G}_r = (\mathcal{V}_r, \mathcal{E}_r)$, where each node $v_r \in \mathcal{V}_r$ denotes a residue and each edge $e_r \in \mathcal{E}_r$ denotes the connection between residues. Utilizing both the protein's one-dimensional sequence and its 3D structural information, we constructed a node feature matrix $\mathcal{X}_r$ and an adjacency matrix $\mathcal{A}_r$ for the residue-level graph. As summarized in Table 2, the node feature matrix $\mathcal{X}_r \in \mathcal{R}^{L_r \times 115}$ integrates seven types of residue descriptors, thus generating in a 115-dimensional feature vector for each residue in the sequence of length L_r . Implementation details of the residue-level encoding strategies are provided in the Supplementary Section S1. In addition, we set 16 Å as the cutoff threshold and calculate the adjacency matrix $\mathcal{A}_r$ according to the node Euclidean distance.

Similarly, the atom-level graph is defined as $\mathcal{G}_a = (\mathcal{V}_a, \mathcal{E}_a)$ where each node $v_a \in \mathcal{V}_a$ represents an atom, and each edge $e_a \in \mathcal{E}_a$ represents the connection between atoms. Following the approach of Jumper et al. [20], atoms within residues are categorized into 37 types. The one-hot encoding scheme is used to construct the node feature matrix

Table 2 Summary of residue-level node features

Features	Shape	Description
Sequence features group		
Amino acid type	[L,20]	One-hot encodings of 20 amino acid residue types
PSSM	[L,20]	Normalized position-specific scoring matrix
HMM	[L,20]	Normalized Hidden Markov Models Matrix
ESM embedding	[L,33]	pLMs embeddings from ESM-2 with 36 layers and 3B parameters
Structural features group		
DSSP	[L,14]	Secondary structure profiles of proteins
resAF	[L,7]	Atomic features of residues
pseudo-position embedding	[L,1]	The 3D coordinates of each residue's side-chain centroid

$\mathcal{X}_a \in \mathcal{R}^{L_a \times 37}$, where each feature vector indicates an atom type. Following previous studies [28], we set the atomic cutoff distance to 2.3 Å to construct the adjacency matrix $\mathcal{A}_a$.

Residue-level graph feature learning with EGNN

In this study, we design the residue-level feature extraction module using E(n) Equivariant Graph Neural Networks (EGNN) [31]. As an advanced variant of traditional GNNs [40], EGNN employs multiple stacked Equivariant Graph Convolutional Layers (EGCL). EGCL aggregates node features while synchronously updating the node coordinate space, thereby preserving E(3) equivariance in Euclidean transformations (such as rotation and translation). In EGCPPIIS, we stacked 4 layers of EGCL to facilitate effective learning of residue-level features. Each EGCL layer takes the node feature matrix and the node coordinate matrix as input, and performs message passing and feature transformation. The initial residue-level coordinate matrix is constructed using the Cartesian coordinates of the side-chain (SC) atom of each query residue. The initial residue-level feature vector has a dimensionality of 115, representing physicochemical and structural descriptors derived from the protein sequence and three-dimensional structure. Therefore, given the node coordinates x_i^l and node features h_i^l of the layer l , each layer of EGCL performs coordinate and node embedding updates through a message passing mechanism, i.e. $x_i^{l+1}, h_i^{l+1} = EGCL[x_i^l, h_i^l]$. The coordinate embedding x_i^{l+1} of node i is updated by the following formula:

$$m_{ij} = \phi_e \left(h_i^l, h_j^l, \|x_i^l - x_j^l\|^2 \right) \quad (1)$$

$$C = \frac{1}{M-1}, m_{ij} \in M \quad (2)$$

$$x_i^{l+1} = x_i^l + C \sum_{j \neq i} (x_i^l - x_j^l) \phi_x(m_{ij}) \quad (3)$$

Here, x_j^l and h_j^l denote the coordinate embedding and node embedding of node j at layer l ; $\|x_i^l - x_j^l\|^2$ is the spatial relative distance between nodes i and j ; ϕ_e and ϕ_x are the MLP nonlinear functions for edge and node transformations, respectively; C is a normalization constant, and M indicates the total number of nodes in the residue-level graph. The coordinate update mechanism ensures equal variance in the message passing process for each EGCL layer. Specifically, the update of node coordinates x_i^{l+1} considers the sum of the weighted differences between node i and all other nodes.

Different from traditional GNNs that aggregate local neighbor information, EGCL uses all the node information in the graph when updating node features, preferring to capture global information. The node embedding h_i^{l+1} of node i is updated by aggregating global messages:

$$m_i = \sum_{j \neq i} m_{ij} \quad (4)$$

$$h_i^{l+1} = \phi_h(h_i^l, m_i) \quad (5)$$

where ϕ_h is the MLP nonlinear operation for node feature update.

Atom-level graph feature learning with GraphSAGE

In EGCPPIS, the atom-level feature extraction module is constructed based on GraphSAGE (Graph SAmple and aggreGatE) [41]. GraphSAGE is an efficient variant of GNNs built for inductive representation learning on large-scale graphs. Unlike traditional GNNs that rely on a transductive paradigm, GraphSAGE enables inductive learning on dynamic graphs in an unsupervised manner, and generalizes effectively to previously unknown nodes. In this work, we stack four SAGEConv layers to capture node embeddings of atom-graphs. SAGEConv layer updates node embeddings by performing sampling and aggregation over local neighborhoods. Specifically, during the neighborhood sampling phase, GraphSAGE selects a fixed-size subset of neighbors $N(v_a)$ for each target node $v_a \in \mathcal{V}_a$ in the atom-level graph $\mathcal{G}_a = (\mathcal{V}_a, \mathcal{E}_a)$. In the aggregation phase, we use the default mean aggregator to fuse the neighborhood features of the target node v_a at layer l to generate the aggregated representation $h_{N(v_a)}^{l+1}$. Formally, the aggregation and node update steps are defined as:

$$h_{N(v_a)}^{l+1} \leftarrow \text{mean}(\{h_u^l, u \in N(v_a)\}) \quad (6)$$

$$h_{v_a}^{l+1} \leftarrow \sigma(W^l \cdot \text{concat}(h_{v_a}^l, h_{N(v_a)}^{l+1}) + b^l) \quad (7)$$

where W^l and b^l denote the learnable weight matrix and bias, $\sigma(\cdot)$ represents the activation function. The aggregated neighborhood features $h_{N(v_a)}^{l+1}$ are combined with the target node v_a 's own features $h_{v_a}^l$ and transformed through a nonlinear mapping to produce updated embeddings.

Molecular representation graph contrastive learning strategy

The hierarchical graph structure represents a protein chain at multiple levels, including both residue-level and atom-level graphs. Based on this, residue-level embeddings capture global features, while atom-level embeddings represent local information. In this work, EGCPPIS uses a GNN-based comparative learning strategy for molecular representation to fuse node embeddings under different hierarchical levels [42–44]. Graph contrastive learning strategy aims to enhance the consistency between hierarchical levels by maximizing mutual information, thus improving the quality of the learned node embeddings. First, residue-level embeddings $h_{V_i}^r$ and atom-level embeddings $h_{V_i}^a$ are projected into a unified feature space using MLP:

$$\bar{h}_{V_i}^r = MLP(h_{V_i}^r) \quad (8)$$

$$\bar{h}_{V_i}^a = MLP(h_{V_i}^a) \quad (9)$$

where MLP function is composed of two feedforward layers, and both levels share the same weight parameters. Subsequently, EGCPPIS uses the contrastive learning strategy to refine the representations at different levels. The performance of contrastive learning relies on the effective design of positive and negative sample pairs. The rule for constructing positive sample pairs is defined as follows: the anchor point representation should be consistent with the positive sample and different from the negative sample. For a given query residue as the anchor, Positive sample pairs are defined as the top 10% most similar residues to the anchor residue, while the remainder form the negative set. Furthermore, we optimize the residue embeddings through the loss function to make it close to the positive sample and far away from the negative sample. Inspired by InfoNCE [45], the residue-level contrastive loss is formulated as:

$$L_{V_i}^r = -\log \frac{\sum_{V_u \in \mathbb{P}_{V_i}} \exp \left(\cos(\bar{h}_{V_i}^r, \bar{h}_{V_u}^a) / \tau \right)}{\sum_{V_k \in (\mathbb{P}_{V_i} \cup \mathbb{N}_{V_i})} \exp \left(\cos(\bar{h}_{V_i}^r, \bar{h}_{V_k}^a) / \tau \right)} \quad (10)$$

where, $\cos(\cdot)$ denotes cosine similarity, τ is the temperature hyperparameter, $\mathbb{P}_{V_i}$ and $\mathbb{N}_{V_i}$ represents the positive and negative sample set of V_i , respectively. Similarly, the loss function based on the atom-level graph is defined as:

$$L_{V_i}^a = -\log \frac{\sum_{V_u \in \mathbb{P}_{V_i}} \exp \left(\cos(\bar{h}_{V_i}^a, \bar{h}_{V_u}^r) / \tau \right)}{\sum_{V_k \in (\mathbb{P}_{V_i} \cup \mathbb{N}_{V_i})} \exp \left(\cos(\bar{h}_{V_i}^a, \bar{h}_{V_k}^r) / \tau \right)} \quad (11)$$

To integrate both levels of representation, we define the final contrastive loss L^{GCL} as follows:

$$L^{GCL} = \frac{1}{|D|} \sum_{i=1}^{|D|} [\lambda \cdot L_{V_i}^r + (1 - \lambda) \cdot L_{V_i}^a] \quad (12)$$

where $|D|$ is the number of anchors and λ is a learnable hyperparameter balancing the two levels.

The rationale behind applying contrastive learning at both residue and atom levels is to ensure consistency and complementarity between hierarchical representations. The atom-level graph captures fine-grained local structural and chemical information (e.g., atomic interactions and bond types), whereas the residue-level graph encodes higher-level contextual and spatial relationships among amino acids within the protein chain. By introducing contrastive learning at both levels, EGCPPIS enforces the embeddings of corresponding residues and their constituent atoms to be mutually informative and consistent in the unified feature space. This hierarchical contrastive mechanism enables the model to learn representations that preserve both global topology and local physicochemical patterns, thus enhancing the discriminative power of molecular features for downstream prediction tasks.

Improved gated multi-head attention

The attention mechanism originates from the Transformer architecture [46]. To improve the model's sensitivity to critical features, we construct the improved gated multi-head attention module, as illustrated in Fig. 3, to reweight the residue embeddings extracted from the hierarchical graph. First, we merge the node embeddings of the two levels to obtain $f \in R^{N \times 230}$, and treat it as a query matrix Q , a key matrix K , and a value matrix V . Since standard self-attention may overemphasize self-representation while neglecting diverse contextual information, we adopt the multi-head attention mechanism for modeling different subspaces in parallel. For each attention head, we compute the attention weights using:

$$A_i = \text{softmax} \left(\frac{(QW_i^Q)(KW_i^K)}{\sqrt{d_k}} \right) \quad (13)$$

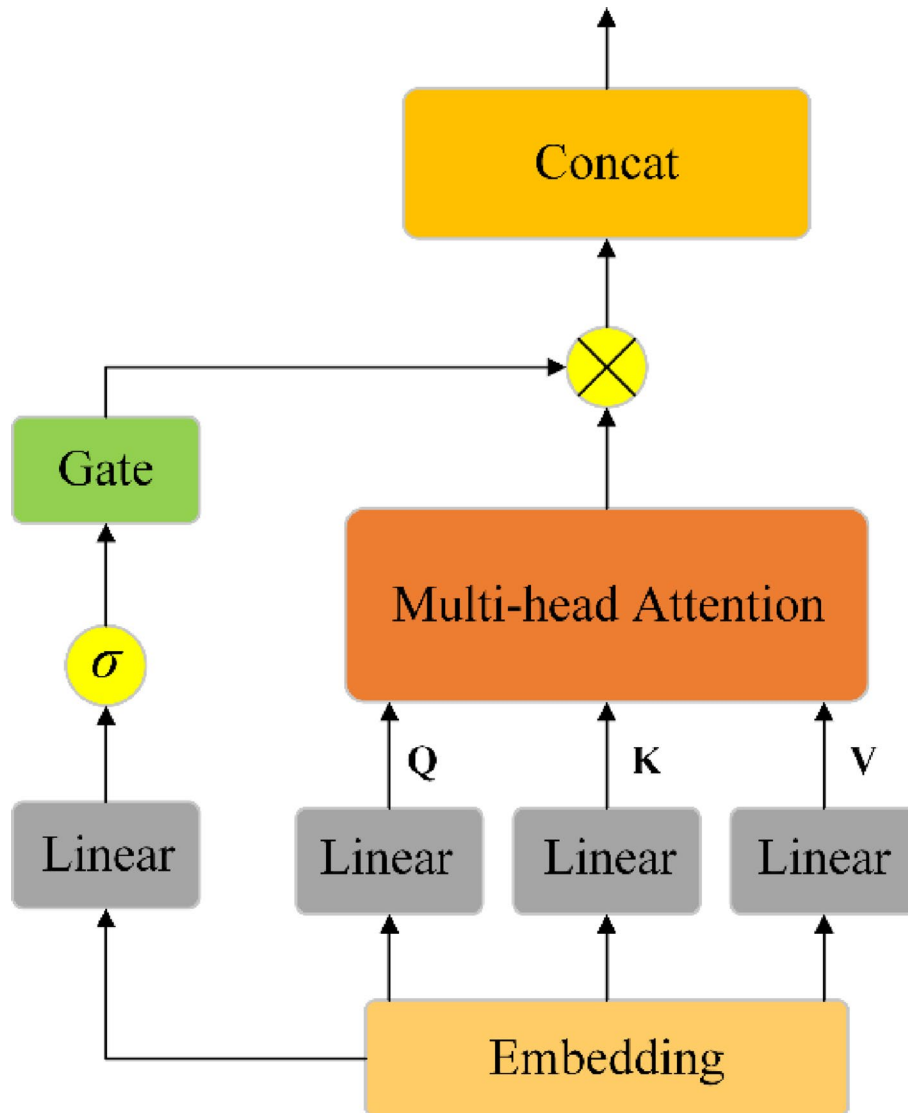


Fig. 3 The framework of improved gated multi-head attention module

$$\text{head}_i = A_i (VW_i^V) \quad (14)$$

where W_i^Q , W_i^K , and W_i^V denote the learnable parameter matrices corresponding to Q , K , and V , respectively. $A_i \in R^{N \times N}$ is the attention matrix, where $i = 1, \dots, H$. EGCPPIs sets the number of attention heads $H = 10$.

To dynamically regulate the output of node embeddings, we incorporate a gated mechanism [47] inspired by LSTM [48], which modulates the multi-head outputs. The final residue embedding output h_{gated} is computed as:

$$G = \sigma(fW^G + b^G) \quad (15)$$

$$h_i^{gated} = G \odot \text{head}_i \quad (16)$$

$$h_{gated} = \text{concat}(h_i^{gated}, \dots, h_H^{gated})W \quad (17)$$

where W^G , b^G , and W are learnable parameters, and $\odot$ denotes element-wise vector multiplication.

Experimental settings

In this work, EGCPPIs uses the binary cross-entropy loss function for PPIs classification:

$$L^{BCE} = \frac{1}{|\Omega|} \sum_{v_i \in \Omega} (y_i \ln \hat{y}_i + (1 - y_i) \ln (1 - \hat{y}_i)) \quad (18)$$

where y_i represents the label of residue and $\hat{y}_i$ represents the predicted probability corresponding to y_i . Ω represents the set of all residue nodes. The final training objective function integrates classification and contrastive losses:

$$Loss = L^{BCE} + \delta L^{GCL} \quad (19)$$

where δ is a balancing coefficient controlling the contribution of the contrastive loss.

During model training, we apply grid search to tune hyperparameters including the number of GNN layers and embedding dimensions. The model is trained using Adam optimizer [49] with an initial learning rate of 0.0008, dynamically adjusted via ReduceLROnPlateau (reduction factor=0.2, patience=5). The Training is conducted over 60 epochs, with each taking approximately 70 s on a single NVIDIA RTX 3090 GPU.

Results and discussion

Evaluation metrics

Following previous studies, we evaluate model performance using seven standard metrics: accuracy (ACC), precision, recall, F1-score (F1), Matthews correlation coefficient (MCC), area under the receiver operating characteristic curve (AUC) and area under the precision-recall curve (AUPR). The formulas for all evaluation metrics are provided in Supplementary Section S2. Given the imbalanced nature of the datasets used in this work, MCC and AUPR, which are sensitive to data imbalance, are considered the core evaluation metrics.

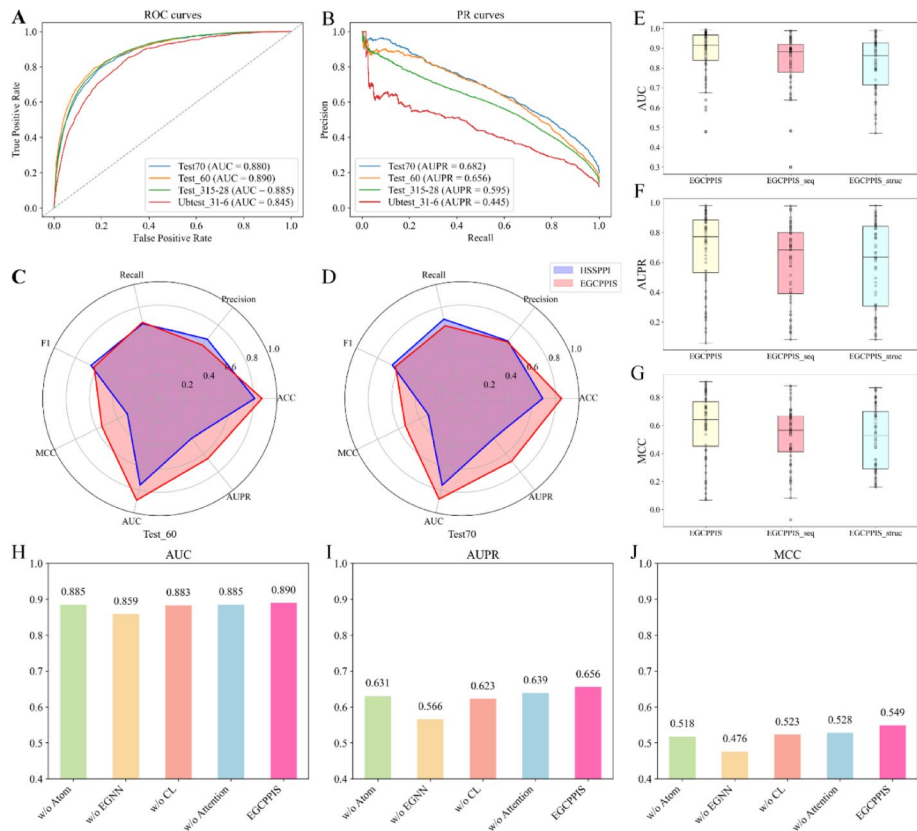


Fig. 4 Comprehensive performance evaluation of EGCPPIS. The ROC curves (A) and PR curves (B) illustrate the performance of EGCPPIS on all test sets. The radar charts (C) and (D) compare the performance of EGCPPIS and the second-best method HSSPPI on the Test_60 and Test70, respectively. The box plots (E), (F), and (G) compare the performance of different variant methods on three core metrics, respectively. The bar charts (H), (I), and (J) compare the performance of individual modules within EGCPPIS on the core metrics

Table 3 Performance comparison of EGCPPIS with existing methods on the Test70

Methods	ACC	Precision	Recall	F1	MCC	AUC	AUPR
DeepPPISP ^b	0.655	0.303	0.577	0.397	0.206	0.671	0.320
DELPHI ^a	0.667	0.320	0.604	0.418	0.236	0.690	N/A
HN-PPISP ^a	0.667	0.324	0.632	0.427	0.244	NA	0.360
EnsemPPIS ^a	0.732	0.375	0.532	0.440	0.277	0.719	0.405
EGRET ^b	0.715	0.358	0.516	0.438	0.270	0.719	0.405
HSSPPI ^b	0.690	0.630	0.696	0.661	0.319	0.759	0.434
EGCPPIS	0.851	0.621	0.639	0.630	0.537	0.880	0.682

^a refers to methods using protein sequence information as input. ^b refers to methods using protein structure

Comparison with state-of-the-art methods on two PPIs prediction tasks

In this work, we conduct comprehensive comparisons between EGCPPIS and state-of-the-art methods on two PPI prediction tasks. Figures 4A and 4B. illustrate the ROC curves and PR curves of EGCPPIS across all datasets, respectively. For the DeepPPISP task, we compare EGCPPIS with six existing methods, including sequence-based models (DELPHI [16], HN-PPISP [17], EnsemPPIS [19]) and structure-based models (DeepPPISP [15], EGRET [26], HSSPPI [28]). The comparison results are summarized in Table 3, showing that EGCPPIS achieves notable improvements over all baselines. As shown in Fig. 4D, EGCPPIS significantly outperforms the second-best method HSSPPI.

Table 4 Performance comparison of EGCPPIIS with existing methods on the Test_60

Methods	ACC	Precision	Recall	F1	MCC	AUC	AUPR
GraphPPIS ^a	0.776	0.368	0.584	0.451	0.333	0.786	0.429
DELPHI ^a	0.697	0.276	0.568	0.372	0.225	0.699	0.319
EnsemPPIS ^a	0.716	N/A	N/A	0.401	0.266	0.726	0.352
AGAT-PPIS ^b	0.856	0.539	0.603	0.569	0.484	0.867	0.574
GACT-PPIS ^b	0.854	0.530	0.646	0.583	0.499	0.876	0.595
HSSPPI ^b	0.811	0.650	0.657	0.653	0.307	0.756	0.434
EGCPPIIS	0.873	0.584	0.668	0.623	0.549	0.890	0.656

^a refers to methods using protein sequence information as input. ^b refers to methods using protein structure information as input

Table 5 Performance comparison of EGCPPIIS with existing methods on the Test_315-28 and Ubtest_31-6

Methods	Test_315-28			Ubtest_31-6		
	MCC	AUC	AUPR	MCC	AUC	AUPR
GraphPPIS ^a	0.432	N/A	N/A	0.327	0.780	0.365
AGAT-PPIS ^b	0.481	0.874	0.572	0.356	0.796	0.368
HSSPPI ^b	0.273	0.743	0.323	0.338	0.800	0.392
EGCPPIIS	0.511	0.885	0.595	0.401	0.845	0.445

^a refers to methods using protein sequence information as input. ^b refers to methods using protein structure information as input

Specifically, EGCPPIIS achieves improvements of 21.8%, 12.1%, and 24.8% over HSSPPI in MCC, AUC, and AUPR, respectively. In addition, EGCPPIIS performed only slightly lower than the HSSPPI on the evaluation metrics precision, recall, and F1, which may be attributed to the high sensitivity of threshold selection.

Then, we explored the performance of EGCPPIIS on the GraphPPIS task. Six existing methods were selected as baselines, including sequence-based methods (GraphPPIS [24], DELPHI [16], EnsemPPIS [19]) and structure-based methods (AGAT-PPIS [27], GACT-PPIS [30], HSSPPI [28]). The performance comparison on Test_60 is recorded in Table 4. EGCPPIIS achieves the best performance on ACC, Recall, MCC, AUC, and AUPR. Figure 4C shows a radar chart comparing EGCPPIIS with HSSPPI, demonstrating the superiority of EGCPPIIS over the suboptimal method. In particular, considering the significant improvement in the core metrics MCC and AUPR, EGCPPIIS outperforms all baseline methods with protein sequence or structure as input.

To verify the generalization ability of EGCPPIIS, we further test it on independent datasets Test_315-28 and Ubtest_31-6. As shown in Table 5, EGCPPIIS consistently outperforms other PPI prediction methods on the core metrics (see Supplementary Table S2 and S3 for details). Notably, EGCPPIIS maintains strong performance even on Ubtest_31-6 containing unbound protein structures. This demonstrates that EGCPPIIS effectively integrates atom-level embedding through the contrastive learning strategy and can fully capture the conformational change information of protein chain. In summary, EGCPPIIS achieves significant performance improvement over existing methods on the imbalanced PPI prediction tasks.

Assessment of the contribution of different residue features

This section evaluates the relative contributions of different residue feature types to the performance of EGCPPIIS. Considering the division rules of existing methods, we design two ablation variants for comparison: EGCPPIIS_seq and EGCPPIIS_struc, while

maintaining identical model architecture. EGCPPIS_seq uses only sequence-based features as input, including one-hot encoding, PSSM, HMM, and ESM-2 embeddings. EGCPPIS_struc uses only structure-based features, including DSSP, resAF, and pseudo-position embeddings. We visualize the performance distributions of the three methods on the Test_60 using boxplots (see Supplementary Table S42 and S5 for details). As shown in Figs. 4E–4G, EGCPPIS_seq and EGCPPIS_struc have comparable performance on three core metrics, but both are significantly lower than the EGCPPIS that integrates sequence and structural features. Specifically, compared to EGCPPIS_seq, EGCPPIS achieves improvements of 4.2% (AUC), 9.5% (AUPR), and 8.9% (MCC). Compared to EGCPPIS_struc, the gains are 4.3% (AUC), 9.4% (AUPR), and 11.8% (MCC). The comparison results demonstrate that both sequence and structural embeddings contribute substantially to PPIs prediction. However, relying on a single modality fails to fully capture the complex characteristics of protein chains. EGCPPIS achieves the superior PPIs prediction accuracy by synergistically utilizing the complementarity of sequence and structural representations.

Effectiveness of basic modules

To evaluate the contribution of individual module within EGCPPIS, we designed the following ablation studies on the Test_60 dataset, keeping all other settings and model components unchanged. The first variant (w/o Atom) removes the GraphSAGE-based atom-level feature encoder, aiming to assess the impact of atom-level representations. Since PPIs prediction is a residue-level task, directly removing the residue-level encoder is not appropriate. Thus, in the second variant (w/o EGNN), we replace the EGNN module with the GraphSAGE encoder consistent with the atom-level graph encoding strategy. The third variant (w/o CL) removes the contrastive learning module, replacing it with direct concatenation of atom-level and residue-level embeddings. The fourth variant (w/o Attention) removes the gated multi-head attention module. Performance comparison between EGCPPIS and other variants is shown in Figs. 4H–4J (see Supplementary Table S6 and S7 for details).

The results demonstrate that each module in EGCPPIS is effective and contributes to the model performance improvement in different degrees. First, Removing the atom-level encoder slightly reduces performance, confirming that atom-level features can effectively assist EGCPPIS for PPIs prediction. Notably, removing the EGNN module results in a substantial performance drop, with all core metrics decreasing by more than 3%. This highlights the superiority of EGNN in capturing structural features while preserving equivariance in the residue-level graph. These results demonstrate the importance of incorporating both atom-level and residue-level features for accurate PPI prediction. Moreover, the w/o CL experiment demonstrates that the contrastive learning strategy effectively models cross-level interactions, thereby enhancing the molecular representation capability of the GNNs. Furthermore, removing the gated multi-head attention mechanism results in a decline of approximately 2% in AUPR and MCC, indicating the necessity of dynamically weighting the fused embeddings. In summary, the individual modules work synergistically, contributing significantly to the competitive performance of EGCPPIS.

Contribution of equivariance

To further verify the superiority of the EGNN module in extracting residue-level equivariant features, we conducted a comparative analysis using several advanced GNNs architectures. To ensure a fair comparison, we kept the input features and all other architectural parameters consistent across models. Specifically, we selected four different paradigms of GNNs to construct the feature extraction module of the residue-level graph in EGCPPIs. Different GNNs architectures use different message passing strategies. ChebConv [50] utilizes Chebyshev polynomial filters, enabling large receptive fields. TransformerConv [51] introduces a Transformer-based attention mechanism. GraphSAGE [41] is an inductive learning framework that generates node embeddings through neighbor sampling and aggregation. GVP-GNN [52] incorporates geometric vector perception, allowing richer handling of vector-valued node features.

Performance comparison of different GNNs models on Test_60 is shown in Fig. 5 (see Supplementary Table S8 for details). Results show that EGCPPIs with EGNN achieves the best performance, outperforming the suboptimal model (GVP-GNN) by over 4% on key metrics AUPR and MCC. The observed improvement is attributed to EGNN's capability to dynamically update node coordinates, enabling more effective encoding of 3D structural information. In summary, the EGNN module can effectively learn residue-level node embedding, thereby enhancing the ability of EGCPPIs to identify PPIs.

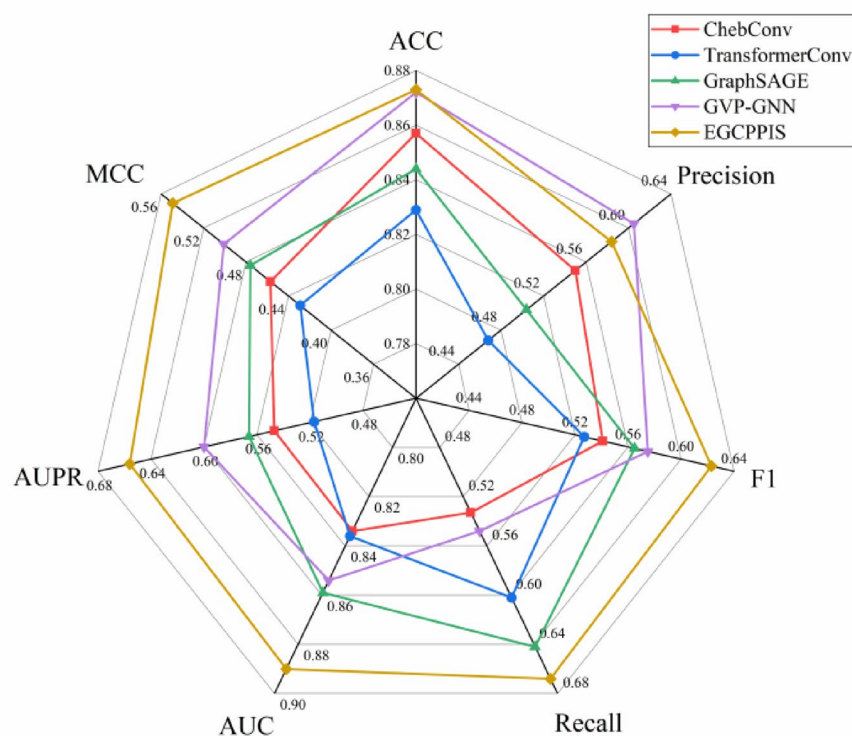


Fig. 5 Performance comparison of different GNNs architectures on Test_60

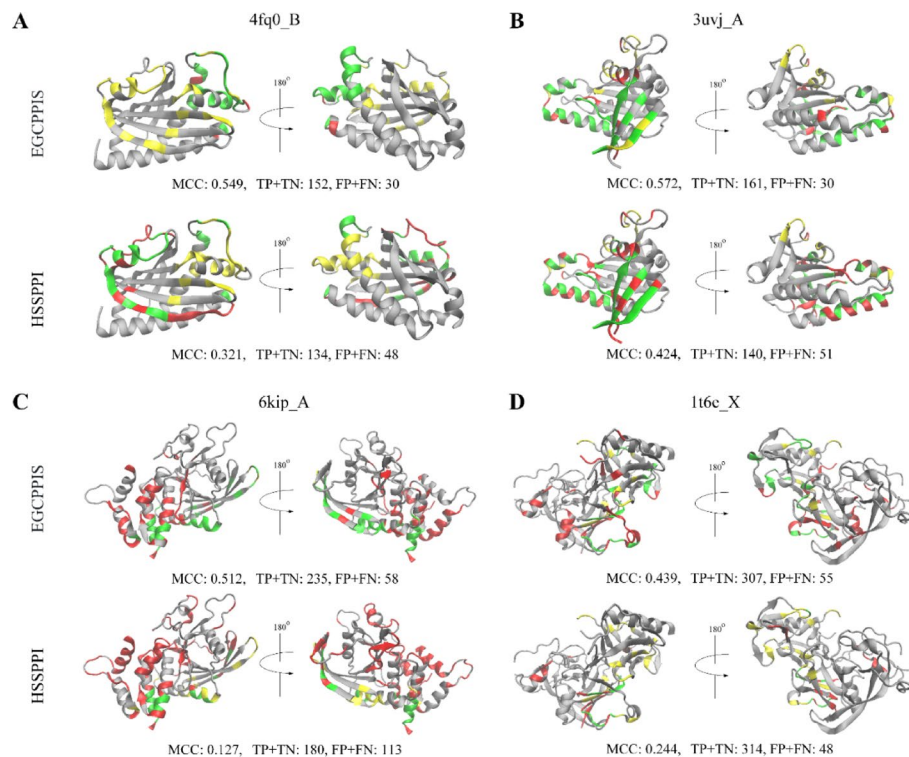


Fig. 6 Representative examples of protein–protein binding site identification generated by EGCPPIS and the second-best method, HSSPPI. Binding sites classified as True Positives (TP), False Positives (FP), and False Negatives (FN) are represented in green, red, and yellow, respectively

Case studies

In this section, we visualized four representative cases predicted by EGCPPIS and the second-best method HSSPPI, selected from four different PPIs test sets. To ensure the reliability of the comparison, we selected the cases with the closest MCC to the overall MCC from each test set. The proteins 4fq0_B, 3uvj_A, 6kip_A, and 1t6e_X were sampled respectively from Test70, Test_60, Test_315-28, and Ubtest_31-6.

As shown in Fig. 6, EGCPPIS displays more prominent green regions (true positives part), whereas HSSPPI exhibits larger red regions (false positives part). Specifically, compared to HSSPPI, EGCPPIS correctly identifies 18, 21, and 55 more residues for 4fq0_B, 3uvj_A, and 6kip_A, respectively. In contrast, for 1t6e_X in Fig. 6D, EGCPPIS yields slightly fewer correct predictions overall. However, it identifies 27 true binding residues, more than twice the number predicted by HSSPPI (11 residues), highlighting EGCPPIS's superior capability in pinpointing true binding sites. Across all cases, EGCPPIS consistently yields higher MCC values, demonstrating enhanced accuracy in identifying both binding and non-binding residues.

Conclusion

Efficient identification of protein–protein interaction sites is critical for understanding biological processes and elucidating disease mechanisms. However, current approaches still face challenges due to limitations in protein feature representation and the design of encoders, leaving potential for further performance enhancement. In this study, we propose EGCPPIS, a novel graph-based learning framework for PPI site prediction. EGCPPIS performs hierarchical molecular representation learning by applying GNNs

to both residue-level and atom-level protein graphs. Contrastive learning strategies and attention mechanisms are then employed to refine the residue embeddings. Finally, the fused embeddings are transferred to MLP for predicting the probability scores of PPIs. The superiority of EGCPPIs is attributed to three key factors. (1) EGCPPIs constructs hierarchical graph representations based on the inherent 3D hierarchical conformation of protein chains to fully extract structural information. (2) EGCPPIs utilizes EGNN as the backbone network, which maintains spatial equivariance throughout feature learning, thereby extracting global features with equivariant perception characteristics in the residue-level graph. (3) EGCPPIs uses the contrastive learning strategy to synergistically optimize consistent features in the hierarchical graph. The residue embeddings are subsequently weighted and updated by an improved gated multi-head attention mechanism for improved representational quality.

EGCPPIs consistently outperformed state-of-the-art baselines on two PPIs prediction tasks. Ablation studies confirmed the contribution of each module to model efficiency, with the EGNN-based residue encoder identified as the core driver of performance improvement. Case studies further demonstrated EGCPPIs's strength in uncovering potential PPI sites.

Despite the excellent performance of EGCPPIs, there is still room for further improvement. The current study focuses on partner-independent PPIs prediction task. In the future, we will consider introducing the characterization of partner proteins, which will predictably further improve the prediction accuracy. Moreover, we aim to extend EGCPPIs to the precise identification of specific functional sites, such as antibody-binding sites and enzyme active sites, to support fine-grained biological insights.

Abbreviations

PPIs	Protein–protein interaction sites
EGNN	E(n) Equivariant Graph Neural Network
CNNs	Convolutional neural networks
pLMs	Protein language models
3D	Three-dimensional
MLP	Multi-layer perceptron
PDB	Protein data bank
EGCL	Equivariant graph convolutional layers
SC	Side-chain
GraphSAGE	Graph sample and aggregate

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12859-025-06328-5>.

Supplementary file1 (DOCX 1206 kb)

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Author contributions

GS proposed and implemented the method, conducted the experiments, and wrote the manuscript. YZ and MZ configured the computational environment, provided optimization insights, and designed the algorithm. YF guided the project, contributed to the conceptualization, and revised the paper. All authors read and approved the manuscript.

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Data availability

The original datasets and code of EGCPPIs are available at <https://github.com/GuicongSun/EGCPPIs>. The repository includes installation instructions, usage examples, and documentation. The DeepPPIsP dataset is available in the Github

repository at <https://github.com/CSUBioGroup/DeepPPISP>. The GraphPPIS dataset is available in the Github repository at <https://github.com/biomed-AI/GraphPPIS>.

Declarations

Ethics approval

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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